

PRODUCT AND CATEGORY DATABASE MANAGEMENT SYSTEM

1. Objective

To design and implement Product and Category tables using Oracle SQL, establish primary and foreign key relationships, perform product insertion, updating and deletion operations, and generate category-wise product reports.

2. Category Table Design

Attribute	Data Type	Constraint	Description
Category_ID	NUMBER	Primary Key	Unique category ID
Category_Name	VARCHAR2(50)	UNIQUE	Name of the category
Description	VARCHAR2(100)	-	Category description

3. Product Table Design

Attribute	Data Type	Constraint	Description
Product_ID	NUMBER	Primary Key	Unique product ID
Product_Name	VARCHAR2(100)	NOT NULL	Name of the product
Category_ID	NUMBER	Foreign Key	References Category table
Price	NUMBER(10,2)	NOT NULL	Price of the product
Stock	NUMBER	NOT NULL	Available stock quantity

4. Primary Key and Foreign Key Relationship

Category_ID is the Primary Key in the **Category** table.

The same Category_ID is used as a Foreign Key in the **Product** table. This relationship connects each product with its corresponding category.

The relationship between Category and Product is **One-to-Many (1)** because one category can contain many products.

5. Create Category Table

```
CREATE TABLE Category (  
    Category_ID NUMBER PRIMARY KEY,  
    Category_Name VARCHAR2(50) UNIQUE,  
    Description VARCHAR2(100)  
);
```

6. Create Product Table

```
CREATE TABLE Product (  
    Product_ID NUMBER PRIMARY KEY,  
    Product_Name VARCHAR2(100) NOT NULL,  
    Category_ID NUMBER,  
    Price NUMBER(10,2) NOT NULL,  
    Stock NUMBER NOT NULL,  
    CONSTRAINT fk_product_category  
    FOREIGN KEY (Category_ID)  
    REFERENCES Category(Category_ID)  
);
```

7. Insert Category Records

```
INSERT INTO Category  
VALUES (1, 'Electronics', 'Electronic products');
```

INSERT INTO Category

VALUES (2, 'Clothing', 'Clothing and fashion products');

INSERT INTO Category

VALUES (3, 'Books', 'Books and educational materials');

INSERT INTO Category

VALUES (4, 'Home Appliances', 'Home and kitchen appliances');

INSERT INTO Category

VALUES (5, 'Beauty', 'Beauty and personal care products');

INSERT INTO Category

VALUES (6, 'Sports', 'Sports and fitness products');

INSERT INTO Category

VALUES (7, 'Toys', 'Toys and games');

INSERT INTO Category

VALUES (8, 'Grocery', 'Food and grocery products');

INSERT INTO Category

VALUES (9, 'Footwear', 'Shoes and footwear products');

INSERT INTO Category

VALUES (10, 'Furniture', 'Home and office furniture');

8. Insert Product Records

INSERT INTO Product

VALUES (101, 'Laptop', 1, 55000, 10);

INSERT INTO Product

VALUES (102, 'Smartphone', 1, 25000, 20);

INSERT INTO Product

VALUES (103, 'T-Shirt', 2, 799, 50);

INSERT INTO Product

VALUES (104, 'Jeans', 2, 1499, 40);

INSERT INTO Product

VALUES (105, 'Python Book', 3, 599, 25);

INSERT INTO Product

VALUES (106, 'Data Science Book', 3, 699, 20);

INSERT INTO Product

VALUES (107, 'Mixer Grinder', 4, 3500, 12);

INSERT INTO Product

VALUES (108, 'Face Wash', 5, 299, 35);

```
INSERT INTO Product
VALUES (109, 'Running Shoes', 9, 2200, 15);
```

```
INSERT INTO Product
VALUES (110, 'Study Table', 10, 4500, 8);
```

9. Product Updating

Update 1 – Change Product Price

```
UPDATE Product
SET Price = 58000
WHERE Product_ID = 101;
```

Update 2 – Change Product Stock

```
UPDATE Product
SET Stock = 30
WHERE Product_ID = 102;
```

Update 3 – Change Price and Stock

```
UPDATE Product
SET Price = 899,
    Stock = 60
WHERE Product_ID = 103;
```

10. Product Deletion

Delete 1

```
DELETE FROM Product
WHERE Product_ID = 108;
```

Delete 2

```
DELETE FROM Product
WHERE Product_ID = 109;
```

Delete 3

```
DELETE FROM Product
WHERE Product_ID = 110;
```

11. Category-wise Product Report

The following query displays products along with their category names.

```
SELECT
    c.Category_Name,
    p.Product_Name,
    p.Price,
    p.Stock
FROM Category c
JOIN Product p
ON c.Category_ID = p.Category_ID
ORDER BY c.Category_Name;
```

12. Conclusion

Thus, the Product and Category database was successfully designed and implemented. The relationship between Category and Product was established using primary and foreign keys. Product insertion, updating, deletion, and category-wise reporting operations were successfully performed using Oracle SQL.